

Demo Video Recording Script — COMP40511 Coursework

Total video length: 10–15 minutes

Before you start recording:

- Turn on your webcam so your face is visible
 - Share your screen showing your Jupyter Notebook (main.ipynb)
 - Make sure your microphone is clear and your room is quiet
 - Speak at a comfortable, steady pace — pause between sections when navigating
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Part 1: Introduction

(1–2 minutes) — Stay on your notebook's opening screen

Say this out loud:

"Hello, my name is [Your Name], and my student ID is [Your ID].

Today I am presenting my coursework for the module COMP40511: Applied Artificial Intelligence.

My project is called 'Apple Leaf Disease Detection and Classification'. The goal of this project is to use four different AI methods to identify diseases in apple leaf images.

The four methods I used are:

- A Multilayer Perceptron — or MLP
- A Random Forest
- A Convolutional Neural Network — or CNN
- K-Means Clustering

The images come from the PlantVillage dataset on Kaggle. There are four types of leaves in the dataset: Apple Scab, Black Rot, Cedar Apple Rust, and Healthy leaves."

Part 2: Project Structure & Data Preparation

(2–3 minutes) — Show your project folder, then scroll to Section 2 and 3 of main.ipynb

Say this out loud:

"Let me quickly show you how the project is set up.

In the project folder, you can see the main folders: `dataset`, `outputs`, and `models`. The `main.ipynb` notebook is where all the work happens.

The dataset is split into three parts: 70% for training, 15% for validation, and 15% for testing.

In Section 2 of the notebook, I load all the images and resize them to 64 by 64 pixels. This makes them smaller and faster to process. I also scale all the pixel values to be between 0 and 1 — this is called normalisation, and it helps the models learn more easily.

For three of the four methods — the MLP, Random Forest, and K-Means — I flatten each image into a single long list of 12,288 numbers.

For the CNN, I keep the images in their original shape of 64 by 64 pixels with 3 colour channels, because the CNN needs to see the full structure of the image.

In Section 3, I visualise the training data. Looking at the class distribution chart, you can see there is an imbalance: there are 1,151 healthy leaf images but only 192 rust images. This is a known limitation.

Also, all images in this dataset have a plain grey background. Real-world photos would have messier backgrounds, so the model may perform differently outside of this dataset."

Part 3: Task 1 — MLP Classifier

(2 minutes) — Scroll to the MLP section in the notebook

Say this out loud:

"Now let's look at Task 1: the Multilayer Perceptron, or MLP.

An MLP is a basic type of neural network. It takes all the input numbers in one go and passes them through several layers to make a prediction.

I built this MLP using Keras. The coursework asked for 4 hidden layers with sizes 15, 25, and 20 neurons. I set the layers as 15, 25, 20, and 15. I trained it using the Adam optimiser for 20 epochs.

The MLP achieved a test accuracy of **64.30%**.

Looking at the learning curves and confusion matrix, you can see the model struggles to tell the diseases apart.

The reason for this is something called a 'bottleneck'. We are feeding in 12,288 pixel values but immediately squeezing them into just 15 neurons. A lot of important visual information gets lost at that step, making it hard for the network to learn the patterns it

needs."

Part 4: Task 2 — Random Forest

(2 minutes) — Scroll to the Random Forest section

Say this out loud:

"For Task 2, I used a Random Forest classifier from the Scikit-Learn library.

A Random Forest works by building many individual decision trees — in this case, 100 trees — and having them all vote on the final prediction. Each tree looks at a random selection of features, and together they are much more accurate than any single tree on its own.

I set a maximum depth of 15 for each tree to prevent overfitting.

The Random Forest achieved a test accuracy of **78.50%** — a big improvement over the MLP.

The reason it works better is that it avoids the bottleneck problem. Instead of compressing everything into a tiny layer first, the trees spread the decision-making across many different pixel combinations. By combining 100 trees, it also becomes much more resistant to overfitting."

Part 5: Task 3 — Convolutional Neural Network and Ablation Study

(3 minutes) — Scroll to the CNN section, show the architecture summary and training curves

Say this out loud:

"For Task 3, I built a Convolutional Neural Network, or CNN.

Unlike the MLP, a CNN is specifically designed to work with images. It uses special layers called convolutional layers to scan across the image and detect patterns like edges, spots, and textures — exactly the kind of features that show up in diseased leaves.

My CNN has exactly 4 convolutional layers, each with 3 by 3 filters, followed by max-pooling layers that reduce the image size step by step.

The main CNN — trained with a learning rate of 0.001 and 50% dropout — achieved a test accuracy of **84.76%**. This is the best result across all four methods.

I also ran an ablation study, which means I changed one thing at a time to see how it affected performance. I reduced the learning rate to 0.0001 and lowered dropout to 30%. This version only achieved **79.33%** accuracy.

Why the drop? A learning rate of 0.0001 is very small. Within only 15 epochs of training, the weights did not have enough time to adjust properly. This shows how important it is to choose the right learning rate — too small and the model does not learn fast enough; too large and it becomes unstable."

Part 6: Task 4 — K-Means Clustering

(2 minutes) — Scroll to the K-Means section, show the Elbow Curve

Say this out loud:

"For Task 4, I used K-Means Clustering. Unlike the other three methods, this is an unsupervised technique — it does not use any labels during training. It simply tries to group similar images together on its own.

To find the right number of clusters, I plotted an Elbow Curve for values of K from 2 to 8. The 'elbow' — the point where the improvement starts to flatten out — appears at K equals 3 and K equals 4. Since we have 4 leaf categories, K=4 makes sense.

However, when I measured how well the clusters matched the real labels, the results were very poor. The Adjusted Rand Index was only 0.0599, and the Normalised Mutual Information was 0.0896. Both scores are close to zero, which means the clusters were barely related to the actual disease categories.

Why does this happen? K-Means groups images based on overall brightness and colour tone. But identifying leaf diseases requires detecting small, specific features like spots and lesions — which K-Means simply cannot do on raw pixel values. This confirms that supervised models like the CNN and Random Forest are essential for this type of task."

Part 7: Conclusion

(1 minute) — Show the final model comparison table at the bottom of the notebook

Say this out loud:

"To wrap up, here is a summary of all four methods side by side.

The CNN is the clear winner with 84.76% accuracy, thanks to its ability to detect spatial patterns in images.

The Random Forest is a strong alternative at 78.50%, without needing the complex architecture of a CNN.

The MLP achieved 64.30%, limited by the information bottleneck in its first layers.

And K-Means Clustering showed that unsupervised methods are not suitable for this kind of fine-grained image classification.

This completes my presentation. Thank you very much for watching."

Recording Tip: Pause naturally between sections while you scroll or navigate. You do not need to rush — a steady, clear delivery is better than a fast one. If you make a small mistake, just continue — minor errors are normal and fine.